

Factory Reallocation & Shipping Optimization Recommendation System for Nassau Candy Distributor

Abstract

Nassau Candy Distributor currently assigns products to factories using static business rules. While this approach simplifies operations, it may result in inefficient shipping routes, higher lead times, and reduced profitability. This project proposes a Factory Reallocation and Shipping Optimization Recommendation System that uses data analytics and decision intelligence to identify better factory-product assignments. The system analyzes product profitability, regional demand, factory performance, and shipping factors to generate actionable recommendations. A Streamlit-based dashboard was developed to enable interactive analysis, what-if simulations, and risk assessment. The proposed solution helps improve operational efficiency while maintaining profitability.

1. Introduction

In modern supply chain management, efficient factory allocation plays a significant role in reducing logistics costs and improving customer satisfaction. Nassau Candy Distributor manufactures and distributes various confectionery products through multiple factories. Existing product-to-factory assignments are based on predefined rules and historical practices rather than data-driven decision making.

As customer demand patterns and regional sales change over time, fixed factory assignments can create inefficiencies. Products may be shipped over longer distances, increasing transportation costs and delivery lead times. Therefore, there is a need for a recommendation system that evaluates factory assignments and identifies opportunities for optimization.

The objective of this project is to analyze historical sales and profitability data and develop an intelligent recommendation system that supports factory reallocation decisions.

2. Problem Statement

Nassau Candy Distributor currently lacks a system capable of:

- Simulating alternative factory-product assignments.
- Measuring the impact of reassignment before implementation.
- Identifying products that should be allocated to different factories.
- Balancing operational efficiency and profitability.

The organization requires a decision-support system that provides data-driven recommendations for improving factory utilization and shipping performance.

3. Dataset Description

The dataset contains 10,194 records and 18 attributes related to orders, customers, products, sales, and profitability.

Key attributes include:

- Order ID
- Order Date
- Ship Date
- Ship Mode
- Region
- Product Name
- Sales
- Units
- Cost
- Gross Profit

Additionally, product-to-factory mappings and factory coordinates were provided to support optimization analysis.

The dataset represents multiple product categories including Chocolate, Sugar, and Other confectionery products.

4. Methodology

4.1 Data Preparation

The dataset was cleaned and transformed using Python and Pandas. Product-to-factory mappings were created based on the provided business rules. Filters and aggregation techniques were applied to analyze profitability and demand patterns.

4.2 Exploratory Data Analysis

Several analyses were performed:

- Factory profitability analysis
- Regional profitability analysis
- Product profitability analysis
- Product demand analysis

These analyses helped identify top-performing factories and products.

4.3 Recommendation Logic

Products generating high profit and demand were classified as strategic products. Low-performing products were flagged for potential reassignment.

A recommendation engine was developed to:

- Rank products based on profitability.
- Evaluate factory performance.
- Suggest alternative factory assignments.
- Estimate projected operational improvements.

4.4 Dashboard Development

A Streamlit dashboard was developed to provide interactive analytics and decision support.

Key modules include:

- Factory Optimization Simulator



Figure 1: Dashboard Overview and Key Performance Indicators (KPIs)

- What-If Scenario Analysis
- Recommendation Dashboard
- Risk and Impact Panel

The dashboard allows users to filter data by region, factory, and shipping mode while evaluating optimization scenarios.

5. Results and Findings

The analysis revealed several important insights.

Factory Performance

Lot's O' Nuts generated the highest overall profit, followed by Wicked Choccy's. These factories contribute the majority of business profitability and should remain primary production centers.



Figure 2: Factory Profitability Analysis

Regional Performance

The Pacific region generated the highest profit among all regions, indicating strong customer demand and market potential.

Product Performance

The most profitable products were:

- Wonka Bar - Scrumdiddlyumptious
- Wonka Bar - Triple Dazzle Caramel
- Wonka Bar - Milk Chocolate
- Wonka Bar - Nutty Crunch Surprise
- Wonka Bar - Fudge Mallows

These products also showed strong demand and should receive priority in production planning.

Recommendations

Based on profitability and demand analysis:

- High-performing products should remain assigned to Lot's O' Nuts and Wicked Choccy's.
- Low-performing products should be reviewed for reassignment opportunities.

- Factory utilization should prioritize regions with stronger demand.
- Future optimization models should incorporate transportation distance and actual lead time calculations.

	Product Name	Factory	Units	Gross Profit	Priority
13	Wonka Bar - Scrumdiddlyumptious	Lot's O' Nuts	7743	19357.5	High
12	Wonka Bar - Triple Dazzle Caramel	Wicked Choccy's	7596	18610.2	High
10	Wonka Bar - Milk Chocolate	Wicked Choccy's	8267	17443.37	High
11	Wonka Bar - Nutty Crunch Surprise	Lot's O' Nuts	6755	16819.95	High
9	Wonka Bar - Fudge Mallows	Lot's O' Nuts	6914	16593.6	High
6	Lickable Wallpaper	Secret Factory	393	3930	Low
14	Wonka Gum	Secret Factory	478	310.7	Low
0	Everlasting Gobstopper	Secret Factory	13	104	Low
4	Kazookles	The Other Factory	371	92.75	Low
3	Hair Toffee	The Other Factory	17	59.5	Low
1	Fizzy Lifting Drinks	Sugar Shack	21	47.25	Low
5	Laffy Taffy	Sugar Shack	27	33.48	Low
8	SweeTARTS	Sugar Shack	41	28.7	Low
7	Nerds	Sugar Shack	10	7	Low
2	Fun Dip	Sugar Shack	8	4.8	Low

Figure 3: Product Reassignment Recommendations

6. Conclusion

This project demonstrates how data analytics can support factory allocation decisions within a supply chain environment. By combining profitability analysis, demand assessment, and recommendation logic, the proposed system provides actionable insights for Nassau Candy Distributor.

The developed Streamlit dashboard enables decision-makers to evaluate factory assignments, compare scenarios, and assess risks before implementing operational changes. The proposed approach improves decision quality and lays the foundation for future predictive optimization systems.

Future Scope

Future enhancements may include:

- Machine Learning-based lead time prediction.
- Route optimization using geographical coordinates.
- Real-time shipment monitoring.
- Advanced factory allocation algorithms.
- Integration with enterprise supply chain systems.

7. References

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- [4] Tableau Documentation. Available at: <https://www.tableau.com/learn/training>
- [5] NumPy Documentation. Available at: <https://numpy.org/doc/>
- [6] Nassau Candy Distributor Dataset (Project Dataset)